

sift $\int_{t_1}^{t_2} x(t) \delta(t - t_0) dt = \int x(t_0); t_1 < t_0 < t_2$ pos

$\bigcirc$; oh m

scaling $\delta(at+b) = \frac{1}{|a|} \delta(t+\frac{b}{a})$ sample $x(t) \delta(t-t_0) = \underline{x(t_0)} \cdot \delta(t-t_0)$

$x[n]$ $x(t)$
period T

$\delta(t)$ $*$ $u(t)$ $u[n]$
 $\delta[n]$ $*$ $\delta(t \rightarrow 0) \Rightarrow \boxed{\infty}$ $\delta[n \rightarrow 0] = \boxed{1}$

$\delta[n]$ $\uparrow$ $\frac{1}{T}$ $\delta(t)$

LECTURE 02

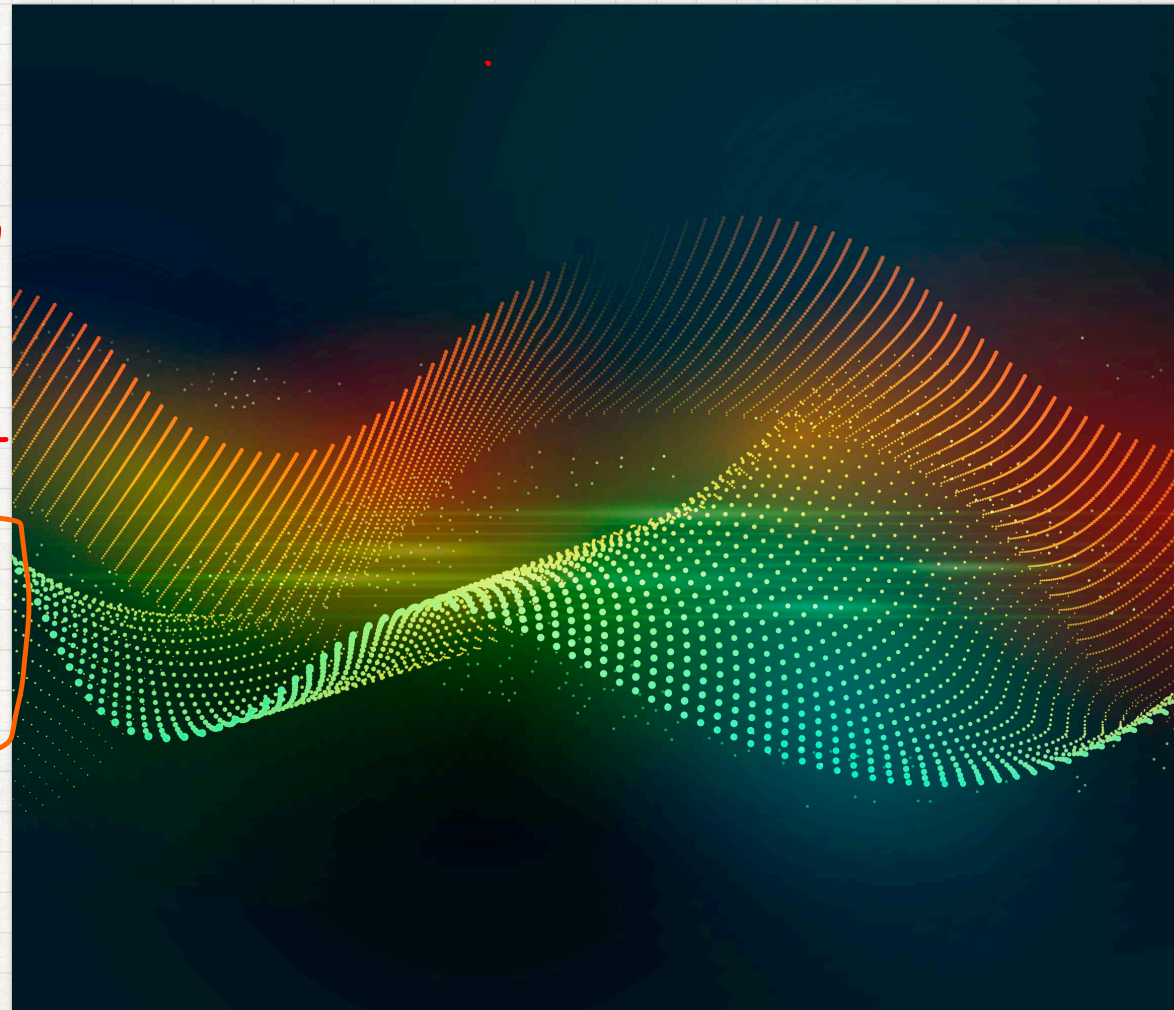
CONVOLUTION

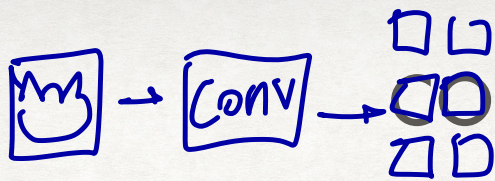
$x[n]$

$\left. \begin{array}{|c|c|c|} \hline 3 & 2 & 1 \\ \hline \end{array} \right\} x[n] = 3\delta[n-1] + 2\delta[n-2] + 1\delta[n-3]$

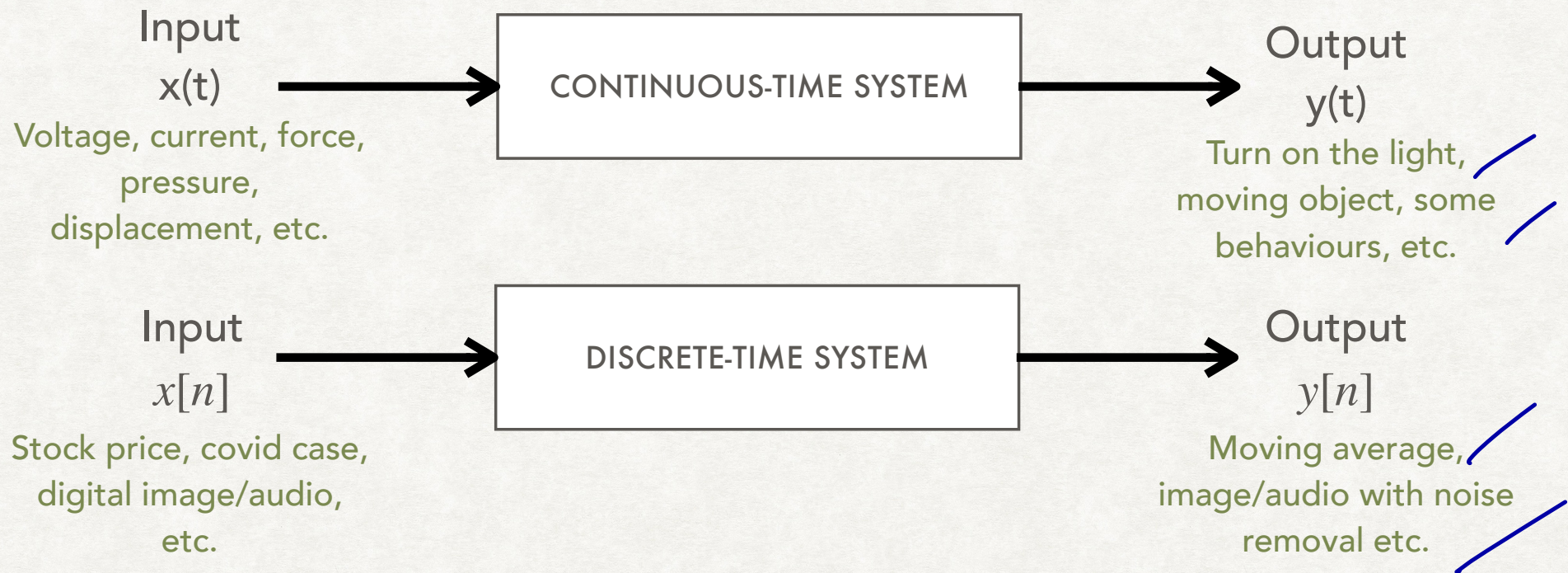
$\left. \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline \end{array} \right\}$

$3\delta[n-1]$ $2\delta[n-2]$ $1\delta[n-3]$





CONTINUOUS & DISCRETE SYSTEMS



Convolutions - Analysis of linear system can be reduced to the study of the system to basic input signals

CLASSIFICATION OF SYSTEMS

System is the process for transforming input resulting in a desired output. There are a number of basic properties of CT and DT systems.

- * • Linear / Non-linear
- * • Time-varying, Time-invariant
- System with/without memory
- Causal system (non-anticipatory)
- Invertibility / Inverse system
- Stable systems (BIBO system)

Basic system properties

1 Linearity



The system is linear if:

- (1) The response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$: additivity properties
- (2) The response to $ax_1(t)$ is $ay_1(t)$, where a is any complex instant : scaling properties

Two properties defining a linear system can be combined into a single statement:

continuous-time: $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$

discrete-time: $ax_1[n] + bx_2[n] \rightarrow ay_1[n] + by_2[n]$

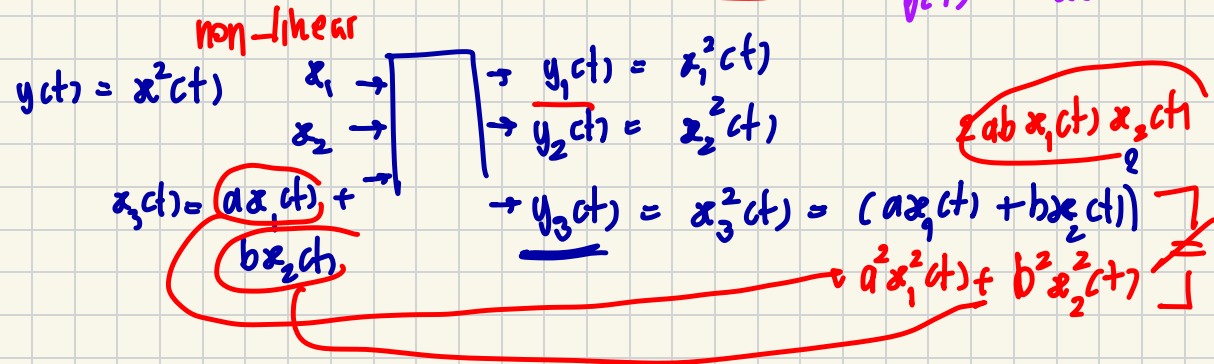
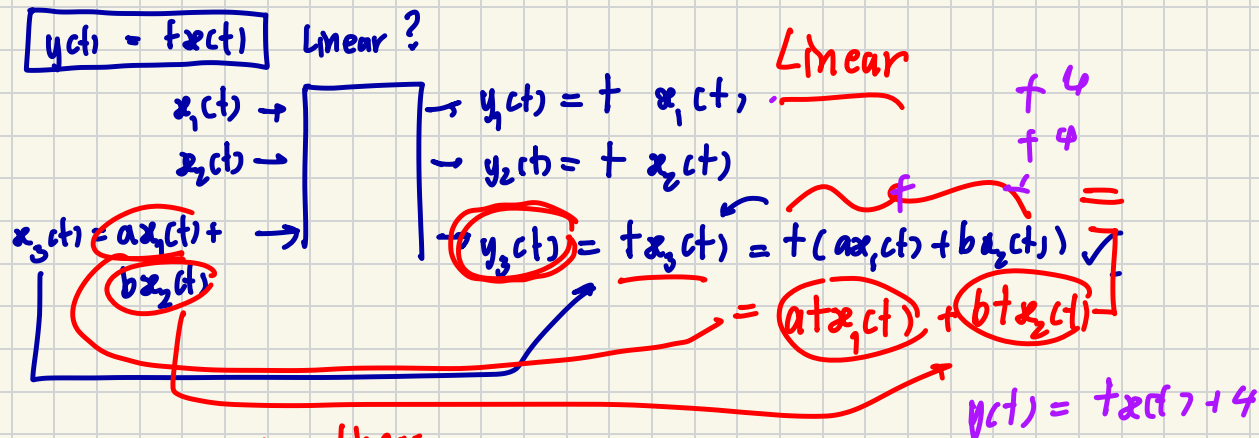
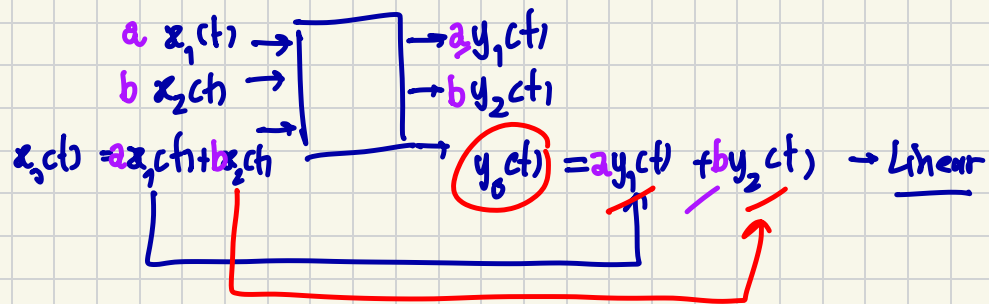
a & b are complex constants

If $x_k[n]$, $k=1,2,3,\dots$ are sets of inputs of a discrete time linear system with the corresponding outputs $y_k[n]$, $k=1,2,3,\dots$, then the response of these inputs given by

is

$$x[n] = \sum_k a_k x_k[n] = a_1 x_1[n] + a_2 x_2[n] + a_3 x_3[n] + \dots$$
$$y[n] = \sum_k a_k y_k[n] = a_1 y_1[n] + a_2 y_2[n] + a_3 y_3[n] + \dots$$

} superposition properties for both continuous & discrete time.



$$1 \quad x_1[n] \rightarrow \boxed{} \rightarrow y_1[n] \quad 1 \quad 2 \quad \neq$$

$$x_2[n] = x_1[n-n_0] \rightarrow \boxed{y_2[n] = y_1[n-n_0]} \rightarrow \text{time-invariant}$$

⑥ $y(t) = \sin(x(t))$

$$x_1(t) \rightarrow \boxed{} \rightarrow y_1(t) = \sin(x_1(t))$$

$$x_2(t) = x_1(t-t_0) \rightarrow y_2(t) = \sin(x_2(t)) = \sin(x_1(t-t_0))$$

$$y_1(t-t_0) = \sin(x_1(t-t_0)) \quad \neq$$

② $y[n] = n \cdot x[n]$

$$x_1[n] \rightarrow \boxed{} \rightarrow y_1[n]$$

$$x_2[n] = x_1[n-n_0] \rightarrow \boxed{y_2[n] = n \cdot x_2[n] = n \cdot x_1[n-n_0]} \quad \text{time-varying}$$

$$y_1[n-n_0] = (n-n_0) \cdot x_1[n-n_0] \quad \neq$$

Ex. System S

$$y(t) = tx(t)$$

[linear] $x_1(t) \rightarrow y_1(t)$

$$y_1(t) = tx_1(t)$$

$$y_2(t) = tx_2(t)$$

$$x_3(t) = ax_1(t) + bx_2(t)$$

$$y_3(t) = atx_1(t) + btx_2(t)$$

$$= ay_1(t) + by_2(t)$$

Ex. $y(t) = x^2(t)$

[non-linear]

$$x_1(t) \rightarrow y_1(t) = x_1^2(t)$$

$$y_2(t) = x_2^2(t)$$

to be linear system:

$$y_3(t) = ay_1(t) + by_2(t)$$

$$y_3(t) = ax_1^2(t) + bx_2^2(t)$$

$$ax_1(t) + bx_2(t) \rightarrow y_3(t)$$

$$y_3(t) = (ax_1(t) + bx_2(t))^2$$

$$(1 \cdot 2 + 0)^2 = \boxed{4}$$

e.g. $x_1(t) = 1$

$x_2(t) = 2, a = 2, b = 0$

not equivalent

$$1^2 \cdot 2 + 0 = \boxed{2}$$

2. Time-invariance

A system is time-invariant if the behavior and characteristics are fixed over time.

A system is time-invariant if a time shift in the input signal results in an identical time shift in the output signal.

$$\begin{aligned}x[n] &\rightarrow y[n] \\ x[n-n_0] &\rightarrow y[n-n_0]\end{aligned}$$

$$\begin{aligned}x(t) &\rightarrow y(t) \\ x(t-t_0) &\rightarrow y(t-t_0)\end{aligned}$$

Ex $y(t) = \sin(x(t))$ [time-invariance]

$$y_1(t) = \sin(x_1(t))$$

$$y_2(t) = \sin(x_2(t))$$

$$\text{if } x_2(t) = x_1(t-t_0)$$

$$y_2(t) = \sin(x_1(t-t_0))$$

$$y_1(t-t_0) = \sin(x_1(t-t_0))$$

also

Ex $y[n] = n x[n]$ [time-varying]

$$\text{if } x_1[n] = \delta[n]$$

$$y_1[n] = n \delta[n] = 0$$

$$\text{if } x_2[n] = \delta[n-1]$$

$$y_2[n] = n \delta[n-1] = 1 \cdot \delta[n-1]$$

$x_2[n]$ is a shifted version of $x_1[n]$

but $y_2[n]$ is not a shifted version

of $y_1[n]$

$\therefore$ time-varying system

3. system with/without memory

A system is memoryless if its output of each value of the independent variable at a given time is dependent only on the input at the same time.

Ex. $y[n] = (2x[n] - x^2[n])^2$ is memoryless ✓ $x(t), x[n]$

$y[n] = \sum_{k=-\infty}^n x[k]$ // accumulator } is a system w/ memory $y[n] = x[n+2]$

4. causal system (non-anticipatory)

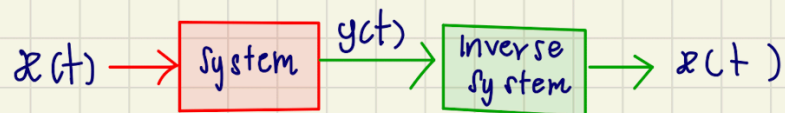
A system is causal if output at any time t_0 depends only on values of input @ $t \leq t_0$

Ex. $\begin{cases} y[n] = x[n] + x[n+1] \\ y(t) = x(t+1) \end{cases}$ ← advanced time } non-causal

All memoryless systems are causal

5. Invertibility (Inverse System)

A system is said to be invertible if distinct inputs leads to distinct outputs.



Ex. $x(t) = \frac{1}{2}y(t)$ ✓

$y(t) = 2x(t) \rightarrow$ inverse $x(t) = \frac{1}{2}y(t)$

$y(t) = \cos(t) \rightarrow$ non-invertible (repeat at $x(t) + 2\pi$)

6. Stable system (BIBO systems) "stability"

Signal $x(t)$ is said to be bounded if its magnitude doesnot grow without band

$|x(t)| < B < \infty$ for all t

A system is BIBO stable (bounded-input bounded output) for any bounded input $x(t)$, the response $y(t)$ is also bounded. $|x(t)| < B_1 < \infty$ implies $|y(t)| < B_2 < \infty$

$u(t), x(t)$



Ex. #1 BIBO $y(t) = e^{x(t)}$ if $|x(t)| < B$, then $|y(t)| = |e^{x(t)}| < e^B < \infty$

#2 $y(t) = \int_{-\infty}^t x(\tau) d\tau$ If $x(t) = u(t)$ $y(t) = \int_{-\infty}^t u(\tau) d\tau = r(t)$

if $|u(t)| < B$, $r(t)$ is unbounded, system is not stable.

References: Section 1.6, A.V. Oppenheim & A.S. Willsky

LINEAR-TIME INVARIANT SYSTEMS (LTI)

✓(1) **superposition property (Linearity)** - the response of the input is given by

$$x[n] = \sum_k a_k x_k[n] = a_1 x_1[n] + a_2 x_2[n] + a_3 x_3[n] + \dots$$

is

$$y[n] = \sum_k a_k y_k[n] = a_1 y_1[n] + a_2 y_2[n] + a_3 y_3[n] + \dots$$

✓(2) **Time - invariance** - the system is time-invariance if a time shift in the input signal results in an identical time shift in the output signal

If

$$x[n] \rightarrow y[n]$$

For time-invariance system, we get

$$x[n - n_0] \rightarrow y[n - n_0]$$

$$\rightarrow h(t) \quad \text{graph of } h(t) \text{ from } -1 \text{ to } \infty$$

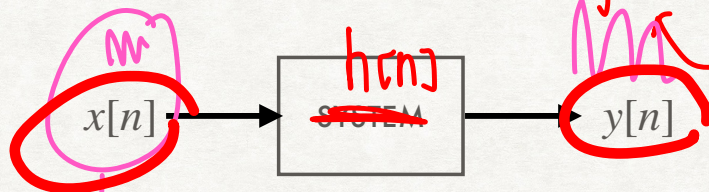
$$u(t) < \infty$$

out

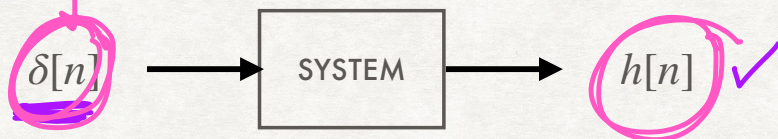
$$e^B$$

Inputs

DISCRETE-TIME SIGNAL CONVOLUTION



Given $h[n]$ is the response of a linear system to the input $\delta[n]$.

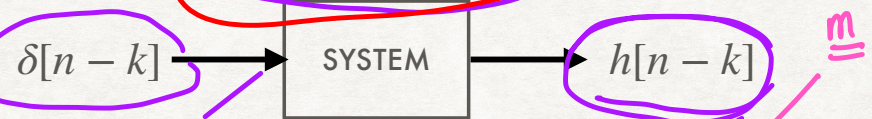


For the discrete-time system, input $x[n]$ can be written as the weighted sum of shifted unit-sampled functions:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

↑
impulse

If the system is time-invariant



From superposition properties,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

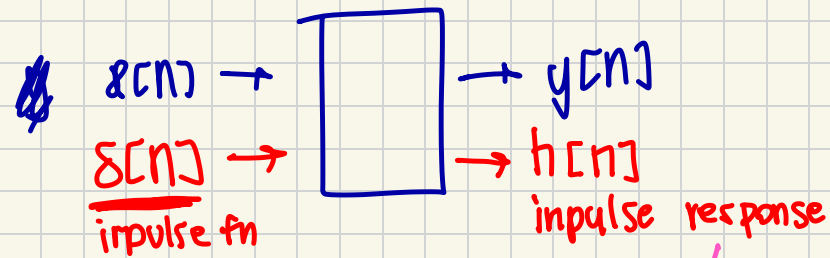
Linear

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k] h[k] = x[n] * h[n]$$

"CONVOLUTION"

$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k] h[k]$$

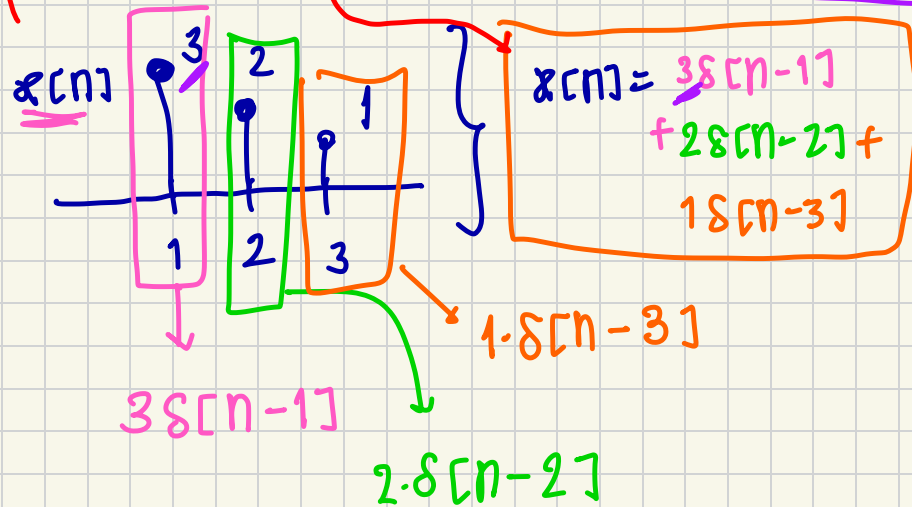
+



$$x[n] = \sum_{k=-\infty}^{\infty} h[k] \delta[n-k]$$

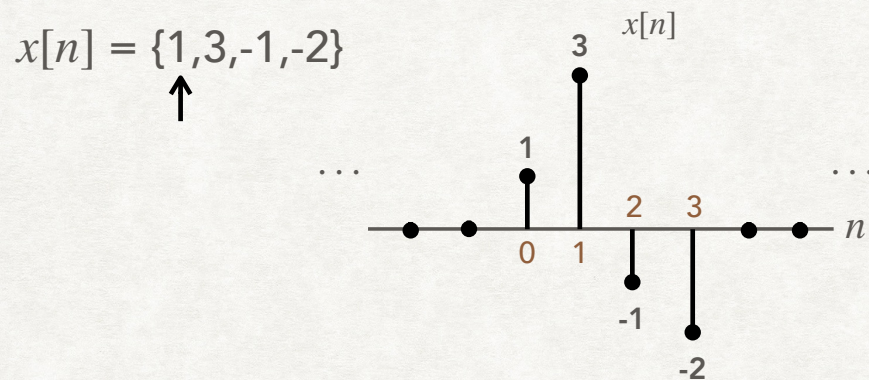
(Note: The original image has a circled $h[k]$ and a pink arrow pointing to it from the text "impulse response".)

impulse response

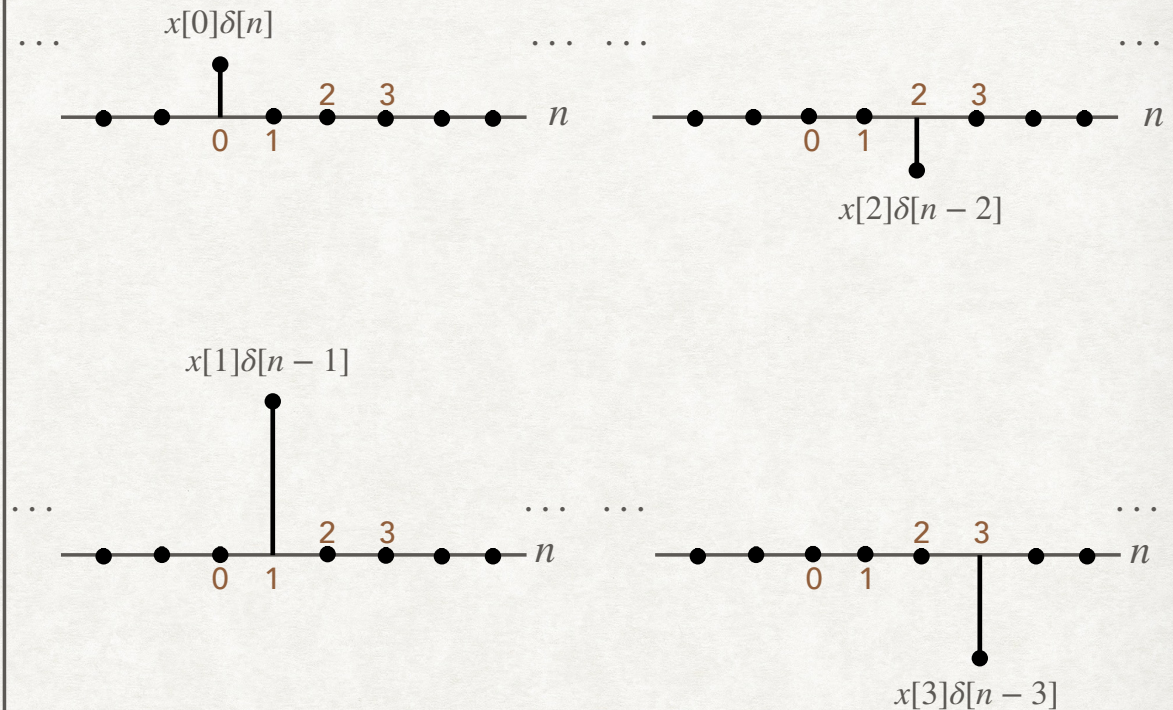


DISCRETE-TIME SIGNAL CONVOLUTION

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$



$$x[n] = x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + x[3]\delta[n-3]$$



Decomposition of a discrete-time signal

GRAPHICAL INTERPRETATION OF CONVOLUTION



EXERCISE 2.1

Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

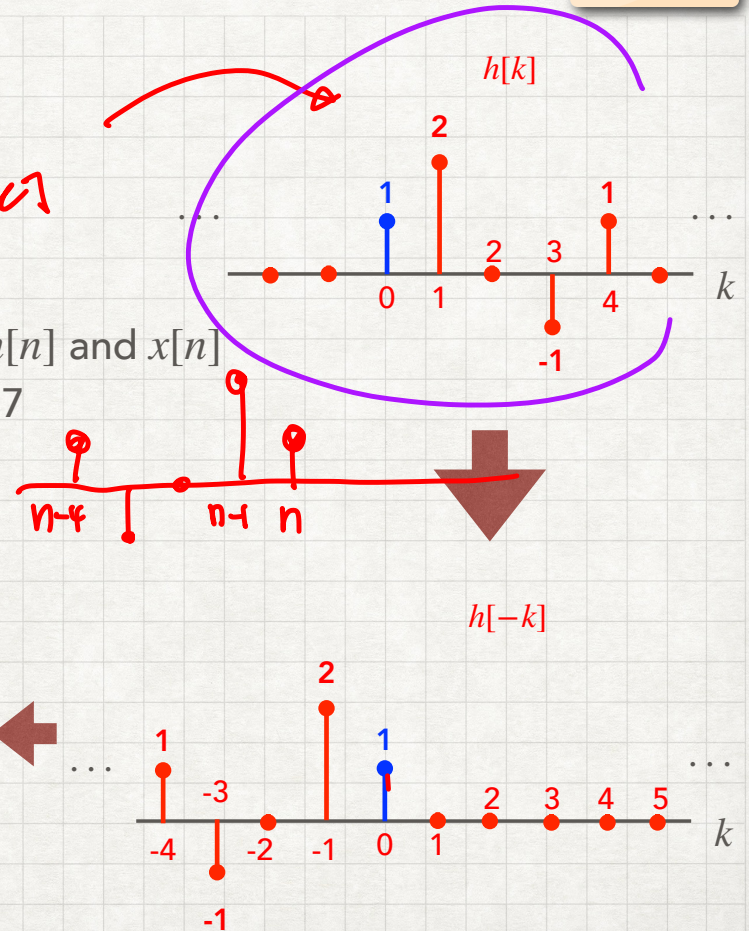
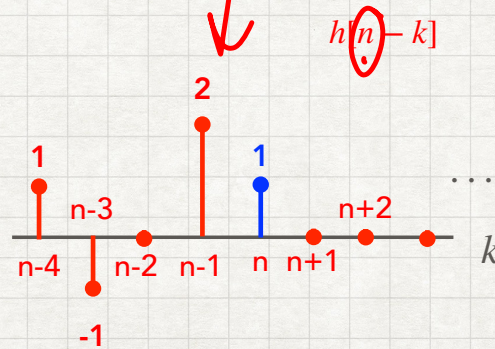
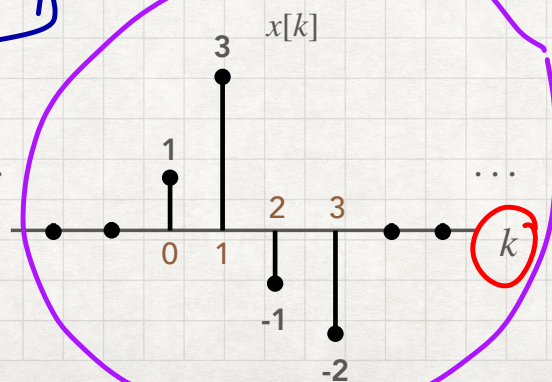
$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

Ans. if M is the length of $h[n]$ and N is the length of $x[n]$
 total points = $M + N - 1$ starting from minimum index between $h[n]$ and $x[n]$
 In this case, total points = $5 + 4 - 1 = 8$ points starting from 0 to 7

Draw $x[k]$ and $h[n-k]$

$(5+4-1)$
 (8)
 input
 9040



GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.1



Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

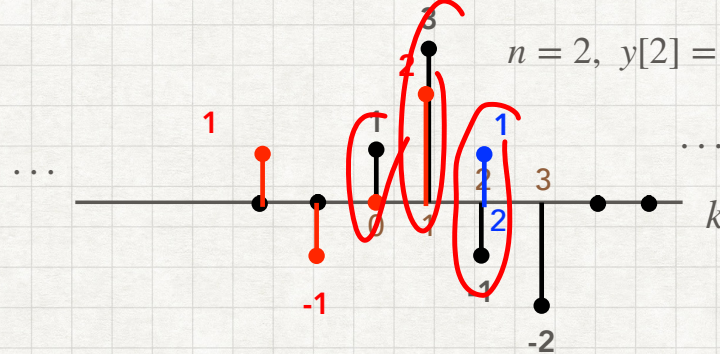
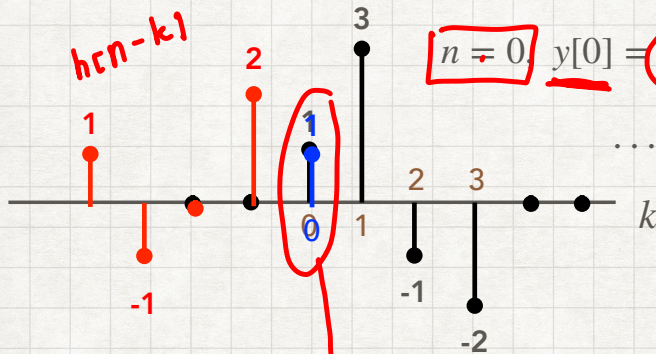
$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$



$h[n-k]$

$n = 0$

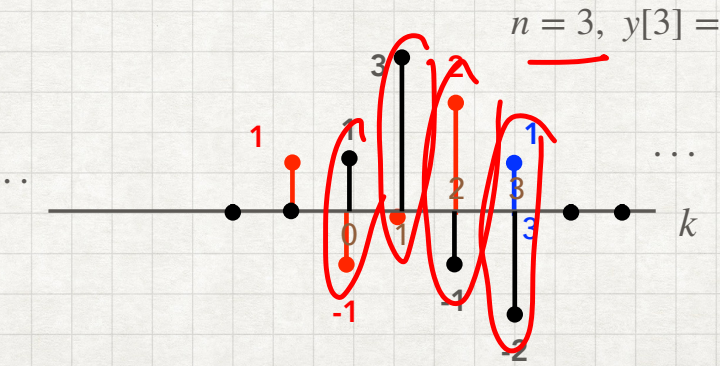
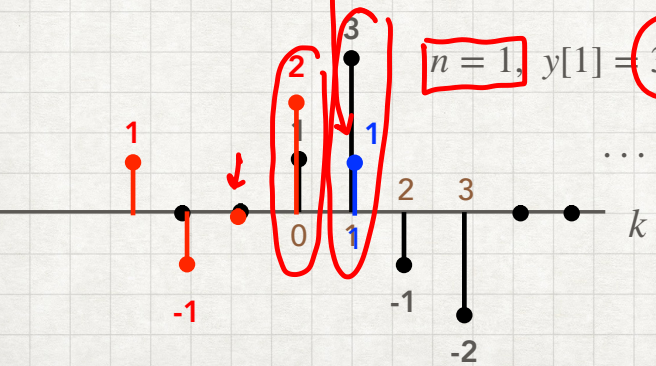
$$y[0] = 1 \cdot 1 = 1$$



$$n = 2, y[2] = -1 \cdot 1 + 3 \cdot 2 + 1 \cdot 0 = 5$$

$n = 1$

$$y[1] = 3 \cdot 1 + 1 \cdot 2 = 5$$



$$n = 3, y[3] = -2 \cdot 1 + (-1 \cdot 2) + 3 \cdot 0 + (1 \cdot -1) = -5$$

GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.1



Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$

$$n = 4, y[4] = -2 \cdot 2 + (-1 \cdot 0) + (3 \cdot -1) + 1 \cdot 1 = -6$$

$$n = 6, y[6] = (-2 \cdot -1) + (-1 \cdot 1) = 1$$

$$n = 5, y[5] = -2 \cdot 0 + (-1 \cdot -1) + 3 \cdot 1 = 4$$

$$n = 7, y[7] = -2 \cdot 1 = -2$$

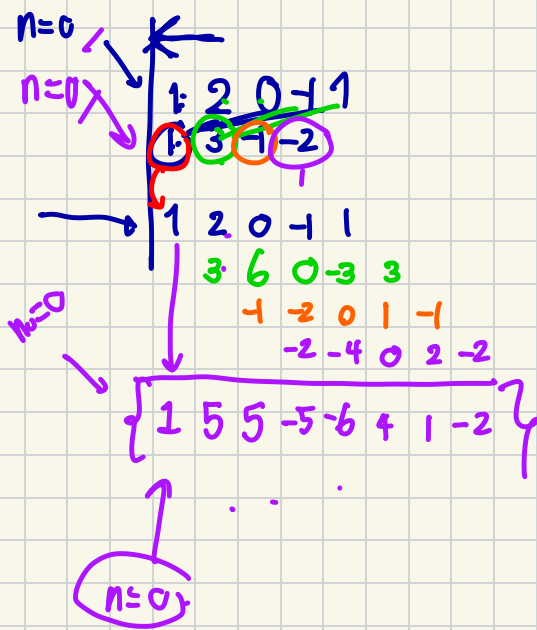
$$y[n] = x[n] * h[n] = \{1, 5, 5, -5, -6, 4, 1, -2\}$$

↑ 54 $n=0$

Ans → {1, 5, 5, -5, -6, 4, 1, -2}

$$\begin{array}{r}
 25 \\
 25 \\
 \hline
 150 \\
 50 \\
 \hline
 \hline
 \end{array}$$

$n=0$



GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.1



Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$

Tabular method

$n=0$ $n=3$ $n=0$

	k	0	1	2	3		
		1	3	-1	-2	<— $x[n]$	
$y[0]$	1	-1	0	2	1	<— Reversed $h[n]$	= 1
$y[1]$		1	-1	0	2	1	= 2+3 = 5
$y[2]$			1	-1	0	2	= 0+6-1 = 5
$y[3]$				1	-1	0	= -1+0-2-2 = -5
$y[4]$					1	-1	= 1-3+0-4 = -6
$y[5]$						1	= 3+1+0 = 4
$y[6]$							= -1+2 = 1
$y[7]$							= -2

$$y[n] = x[n] * h[n] = \{1, 5, 5, -5, -6, 4, 1, -2\}$$



GRAPHICAL INTERPRETATION OF CONVOLUTION



$$\alpha = \frac{1}{2} \quad \beta = \frac{1}{3}$$

EXERCISE 2.2

Let input $x[n] = \alpha^n u[n]$ and impulse response $h[n] = \beta^n u[n]$, Find an output, $y[n]$

$x[k] = \left(\frac{1}{2}\right)^k u[k]$
 $h[k] = \left(\frac{1}{3}\right)^k u[k]$
 $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$

$h[n-k] = \left(\frac{1}{3}\right)^{n-k} u[n-k]$
 $y[n] = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k \left(\frac{1}{3}\right)^{n-k} u[n-k]$
 $y[n] = \left(\frac{1}{3}\right)^n \sum_{k=0}^n \left(\frac{1}{2}\right)^k \cdot 1$
 $y[n] = \left(\frac{1}{3}\right)^n \cdot \left(1 + \frac{1}{2} + \frac{1}{4} + \dots + \left(\frac{1}{2}\right)^n\right)$
 $y[n] = \left(\frac{1}{3}\right)^n \cdot \left(1 - \left(\frac{1}{2}\right)^{n+1}\right) \cdot \frac{1}{1 - \frac{1}{2}}$
 $y[n] = \frac{\left(\frac{1}{3}\right)^n \cdot \left(1 - \left(\frac{1}{2}\right)^{n+1}\right)}{1 - \frac{1}{2}}$
 $y[n] = \frac{\left(\frac{1}{3}\right)^n \cdot \left(1 - \left(\frac{1}{2}\right)^{n+1}\right)}{\frac{1}{2}}$
 $y[n] = 2 \cdot \left(\frac{1}{3}\right)^n \cdot \left(1 - \left(\frac{1}{2}\right)^{n+1}\right)$

Ans. $y[n] = \frac{\beta^n (1 - (\frac{\alpha}{\beta})^{n+1})}{1 - \frac{\alpha}{\beta}} u[n]$

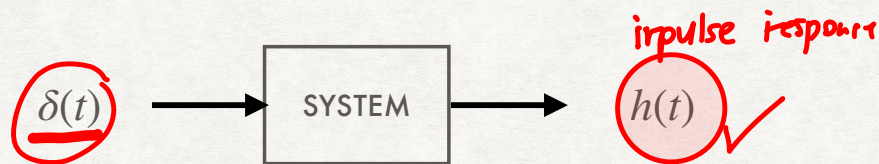
56

14:42

CONTINUOUS-TIME SIGNAL CONVOLUTION



Given $h(t)$ is the response of a linear system to the input $\delta(t)$.



For the continuous-time system, input $x(t)$ can be written as the weighted sum of shifted unite-sampled functions:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

Handwritten notes: A red circle around the integral, an arrow pointing to τ with the word 'integral' below it, and a red scribble to the right.

If the system is time-invariance,



From superposition properties,

$$y(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \quad \rightarrow \quad y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

Handwritten notes: A large red circle around the first integral, and a red circle around the second integral with the text $\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$ written above it.

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau = x(t) * h(t)$$

"CONVOLUTION"

$$h(t) * x(t) = x(t) * h(t)$$

$$\int \delta(\tau) h(t-\tau) d\tau$$

DS / (J) $2\delta(t) * \square = \square$

$\delta(t) * h(t) = h(t)$
 $\delta(t) \rightarrow \boxed{h(t)} \rightarrow h(t)$

CONVOLUTION PROPERTIES

CT convolution satisfies the following important properties

- Commutativity:

$$x(t) * h(t) = h(t) * x(t) \quad \checkmark$$

- Associativity :

$$x(t) * h_1(t) * h_2(t) = [x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)] \quad \checkmark$$

$y(t) = (x(t) * h_1(t)) * h_2(t)$

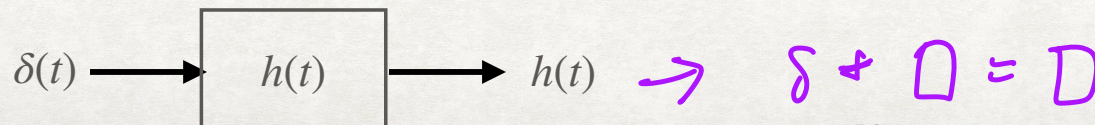
- Distributivity :

$$x(t) * [h_1(t) + h_2(t)] = [x(t) * h_1(t)] + [x(t) * h_2(t)] \quad \checkmark$$

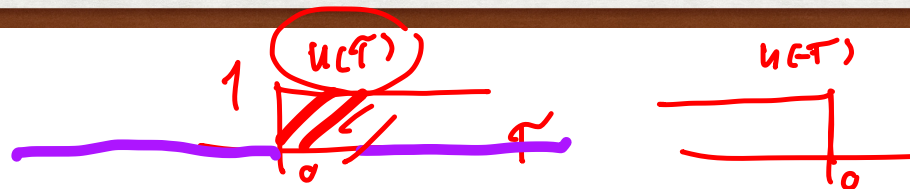
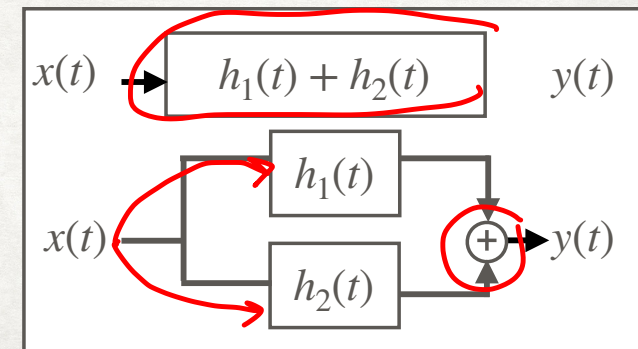
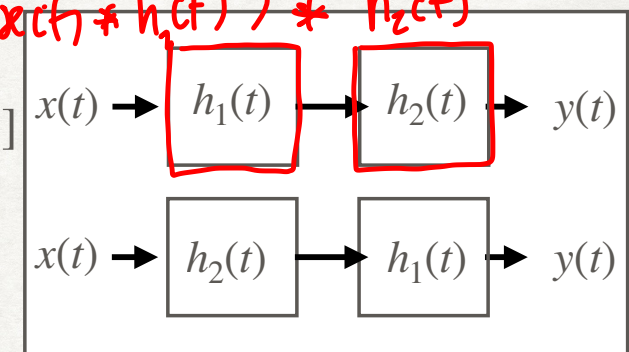
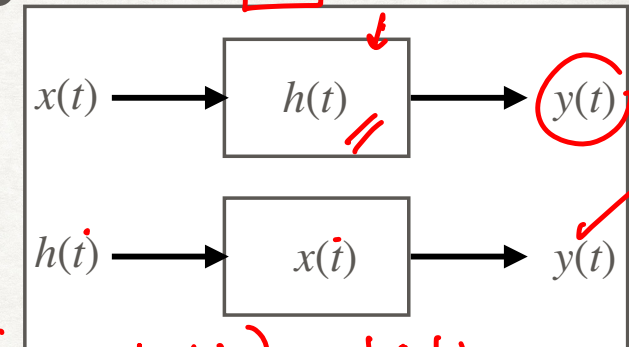
An LTI system with impulse response $h(t) = \delta(t)$ is the identity system

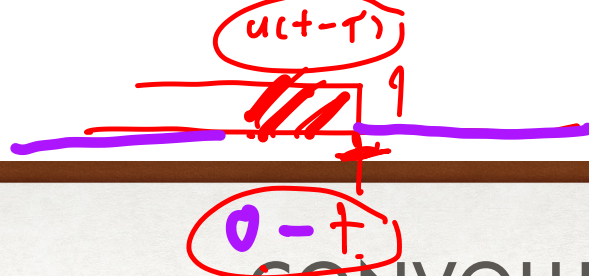
$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t-\tau) d\tau = \int_{-\infty}^t x(\tau) d\tau$$

Also applied for discrete-time convolution.



58





CONVOLUTION EXAMPLE

EXERCISE 2.3



Consider the LTI system with impulse response $\underline{h(t)} = e^{-at}u(t), a > 0$

Input to the system is $\underline{x(t)} = e^{-bt}u(t), b \neq a, b > 0$

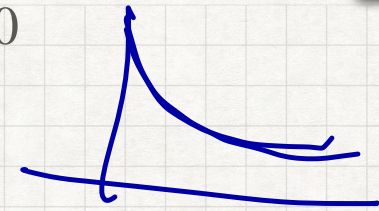
$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \quad \boxed{\text{+ good, } \tau \text{ small}}$$

$$= \int_{-\infty}^{\infty} e^{-b\tau} \underline{u(\tau)} e^{-a(t-\tau)} \underline{u(t-\tau)} d\tau$$

$$= e^{-at} \int_{-\infty}^{\infty} e^{(-b+a)\tau} d\tau \quad t \geq 0$$

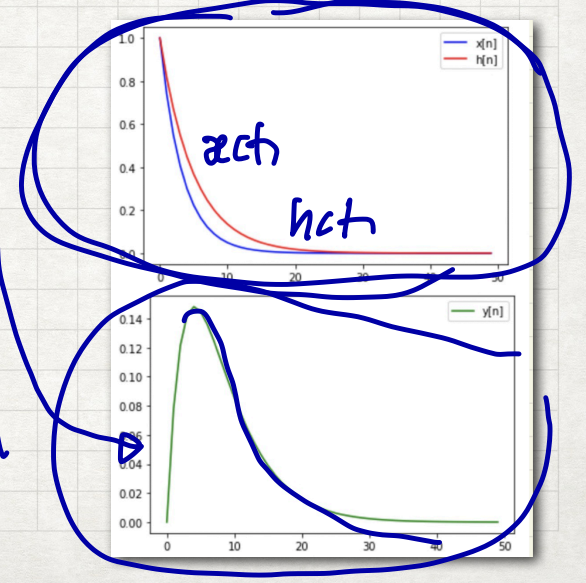
$$= e^{-at} \left(\frac{e^{(-b+a)\tau}}{-b+a} \right) \Big|_{\tau=0}^{\infty} \quad t \geq 0$$

$$= \frac{e^{-at}}{-b+a} (e^{(-b+a)t} - 1), \quad t \geq 0$$



$$y(t) = \frac{1}{a-b} (e^{-bt} - e^{-at}) u(t)$$

If $a = 3, b = 2,$

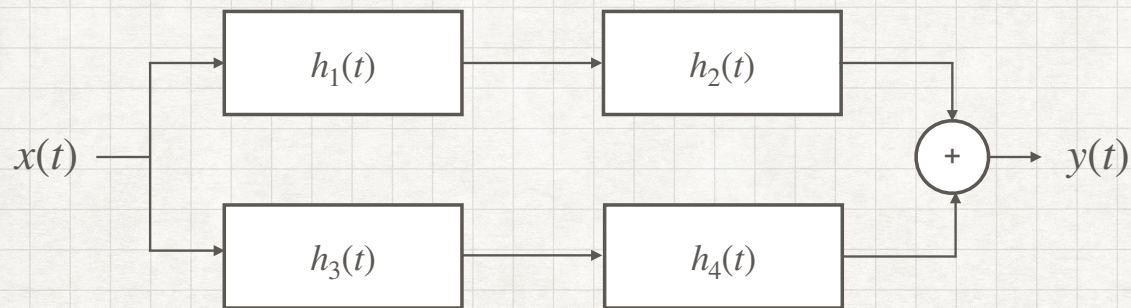


CONVOLUTION EXAMPLE

EXERCISE 2.4



Find the impulse response of the system show in the figure below if



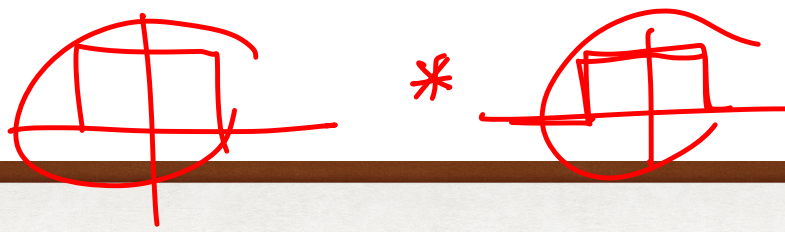
$$h_1(t) = e^{-2t}u(t)$$

$$h_2(t) = 2e^{-t}u(t)$$

$$h_3(t) = e^{-3t}u(t)$$

$$h_4(t) = 4\delta(t)$$

$$(2e^{-2t} - 2e^{-t} + 4e^{-3t})u(t)$$



GRAPHICAL CONVOLUTION INTEGRATION

- When two signals are continuous for all t , convolution is just an ordinary integration. However, when one/both of the signals defined a piecewise fashion, we can use a graphical interpretation of convolution.
- To compute graphical convolution integration in the interval $t_{i-1} \leq t \leq t_i$
 - **Step 1:** fixed value of t in $[t_{i-1}, t_i]$ and **plot $x(t)$ and $h(t - \tau)$ and the product $g(t, \tau) = x(\tau) h(t - \tau)$ as a function of τ .**
 - **Step 2:** **integrate product $g(t, \tau)$** as a function of τ (the integration is the area under the curve). Note that the integral $g(t, \tau)$ depends on t and τ . But **τ will disappear** when the integration is completed.

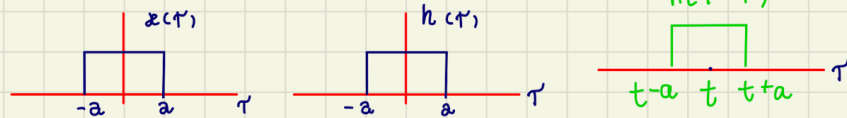
GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.5



Ex VII Determine the convolution

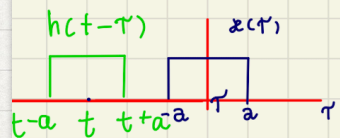
$$y(t) = \text{rect}(t/2a) * \text{rect}(t/2a)$$



How many intervals?

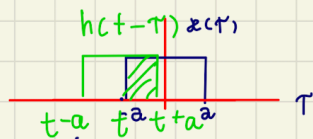
4 intervals

in 1, $t < -2a$, $t \geq 2a$



$$y(t) = 0$$

in 2, $-2a \leq t < 0$



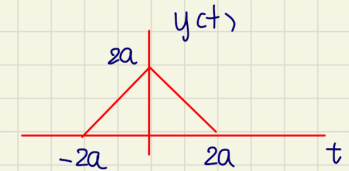
in 3, $0 \leq t < 2a$



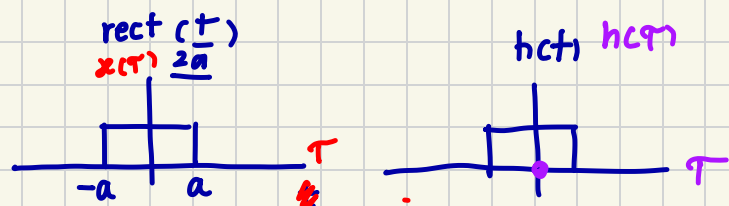
$$\int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-a}^{t+a} 1 d\tau = \tau \Big|_{\tau=-a}^{\tau=t+a} = t+a-(-a) = t+2a$$

$$\int_{t-a}^a 1 d\tau = \tau \Big|_{\tau=t-a}^{\tau=a} = a-t+a = 2a-t$$

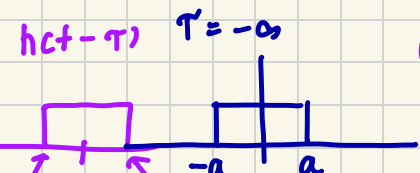
$$y(t) = \begin{cases} 0, & t < -2a \\ t+2a, & -2a \leq t < 0 \\ 2a-t, & 0 \leq t < 2a \\ 0, & t \geq 2a \end{cases}$$



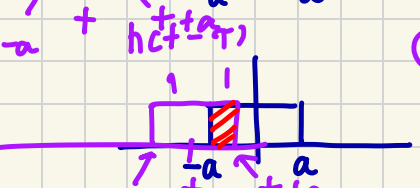
$$y(t) = \begin{cases} 2a-t, & |t| < 2a \\ 0, & \text{otherwise} \end{cases}$$



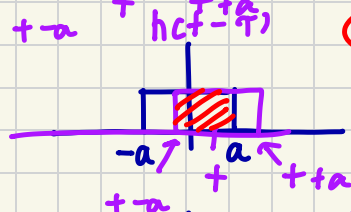
$$y(t) = \int_{-\infty}^{\infty} \underbrace{x(\tau)}_{\text{fixed}} \underbrace{h(t-\tau)}_{\text{slide}} d\tau$$



① $t < -2a$;
 $y(t) = 0$



② $-2a \leq t < 0$
 $y(t) = \int_{-a}^{t+a} 1 \cdot 1 d\tau = \tau \Big|_{-a}^{t+a} = t+2a$

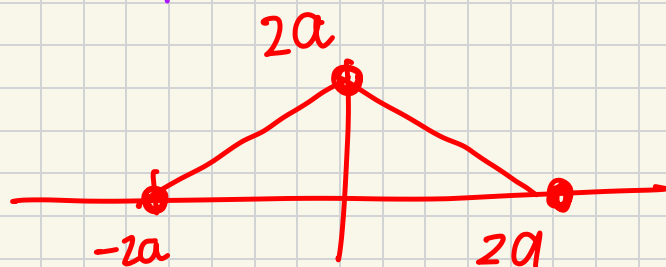


③ $0 \leq t < 2a$
 $y(t) = \int_{t-a}^a 1 \cdot 1 d\tau = a - t + a = -t + 2a$

$$y(t) = \begin{cases} 0 & t < -2a \\ t+2a & -2a \leq t < 0 \\ -t+2a & 0 \leq t < 2a \\ 0 & t \geq 2a \end{cases}$$

$$y(t) = \begin{cases} -|t| + 2a, & |t| \leq 2a \\ 0, & \text{otherwise} \end{cases}$$

④ $t \geq 2a$
 $y(t) = 0$



GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.6



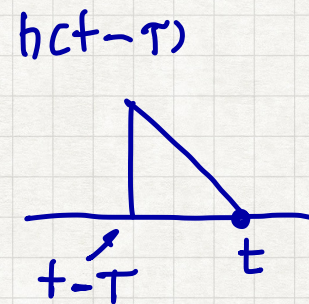
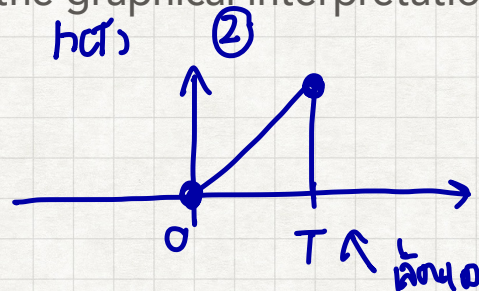
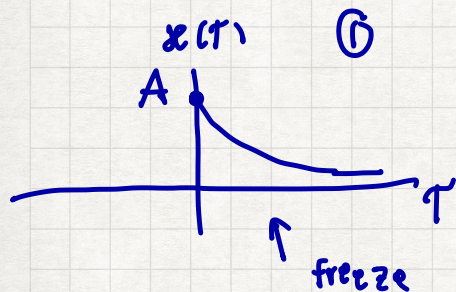
Consider the signals where

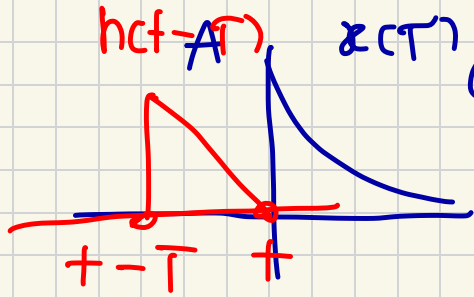
$$x(t) = Ae^{-t}, \quad 0 \leq t < \infty$$

$$h(t) = \frac{t}{T}, \quad 0 \leq t < T$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

Find convolution using the graphical interpretation





① $t < 0 : y(t) = 0$

$$\int e^{-\tau} \tau d\tau = -\tau e^{-\tau} + \int e^{-\tau} d\tau$$

$$\int u dv = uv - \int v du$$

$$\boxed{u = \tau}$$

$$\boxed{du = d\tau}$$

$$dv = e^{-\tau} d\tau$$

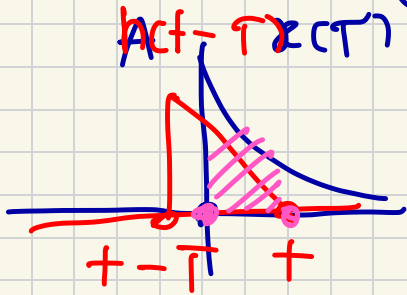
$$\boxed{v = -e^{-\tau}}$$

$$= \left[-\tau e^{-\tau} - e^{-\tau} \right]$$

$0 \leq t < T :$

② $y(t) = \int_0^t A e^{-\tau} \cdot \frac{(t-\tau)}{T} d\tau$

$$= \frac{A}{T} \int_0^t e^{-\tau} t - \underline{e^{-\tau} \tau} d\tau = \frac{A}{T} \left(-e^{-\tau} t + \tau e^{-\tau} + e^{-\tau} \right)_0^t$$



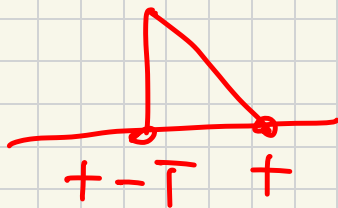
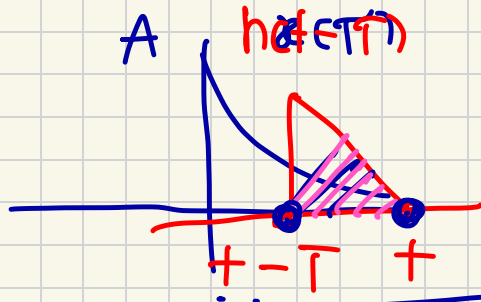
② $t \geq T :$

$y(t) = \int_0^t A e^{-\tau} \cdot \frac{(t-\tau)}{T} d\tau$

$$= \frac{A}{T} e^{-(t-T)} (e^{t-T} + T - 1)$$

$$= \frac{A}{T} \left(-e^{-t} t + t e^{-t} + e^{-t} \right)_0^t$$

$$= \frac{A}{T} (e^{-t} + t - 1)$$

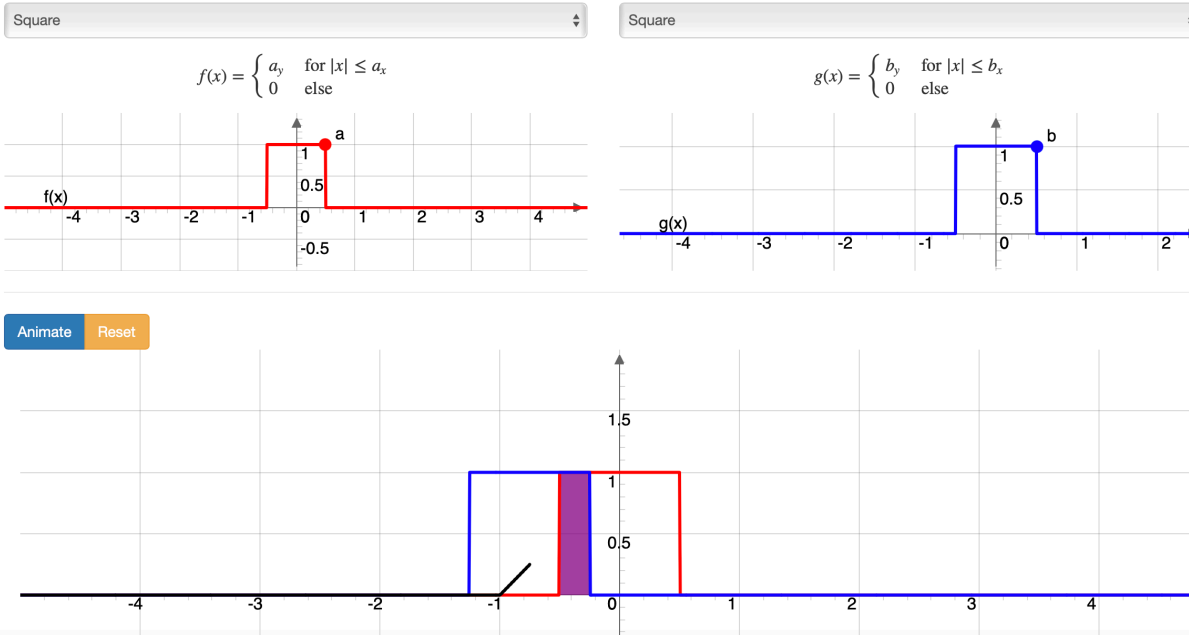


$y(t) = \begin{cases} \text{---} \\ \text{---} \\ \text{---} \end{cases}$

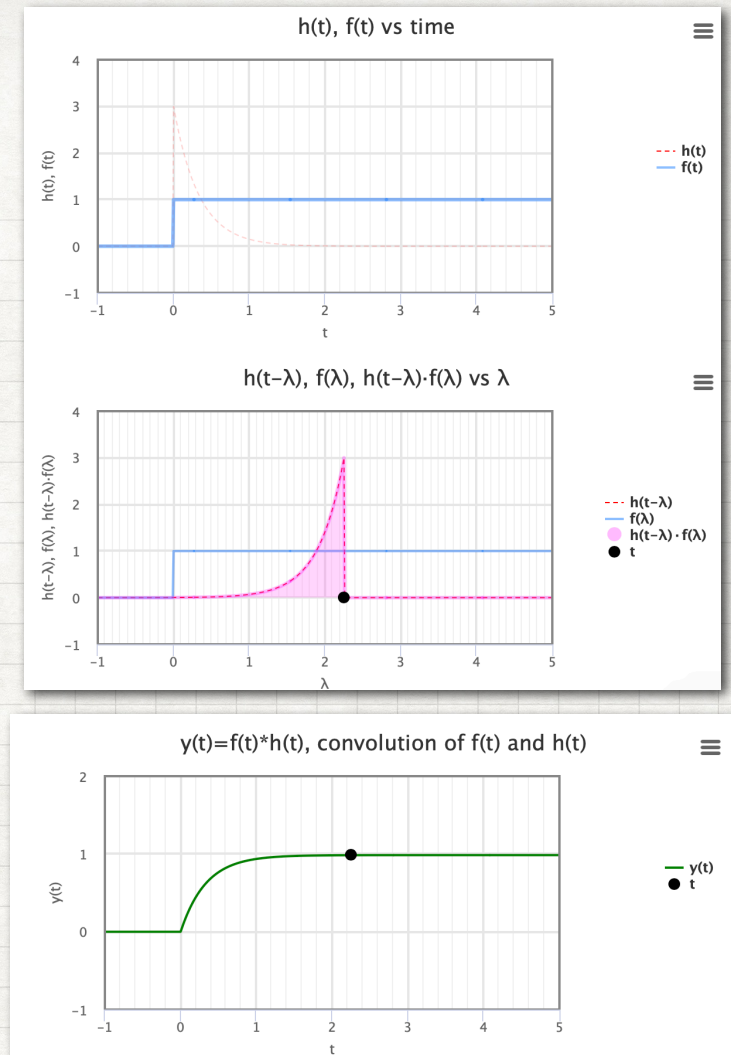
$t <$

DEMO

Convolution demo



<https://phiresky.github.io/convolution-demo/>



<https://lpsa.swarthmore.edu/Convolution/CI.html>

REFERENCES

- Samir S. Soliman and Mandyam D. Srinath., **Continuous and Discrete Signals and Systems**, 2nd Edition, Prentice Hall (1998)
 - Chapter 2 Continuous Systems (2.1,2.2, 2.3) - The convolution Integral, Graphical Interpretation of Convolution
 - Chapter 6 (6.3) Discrete-time Systems (convolution)
- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems**, 2nd Edition, Prentice Hall (1996)
 - Chapter 2 LTI systems - discrete-time and continuous-time convolution integrals

2 (1)

$$\begin{array}{r} 1 \quad 3 \quad 1 \\ 1 \quad 3 \quad 1 \\ 1 \quad 3 \quad 1 \\ 3 \quad 9 \quad 3 \\ \hline 1 \quad 3 \quad 1 \\ 1 \quad 6 \quad 11 \quad 6 \quad 1 \end{array}$$

(a)

2 (2)

$$\begin{array}{r} 3 \quad 2 \quad 1 \\ 3 \quad 2 \quad 1 \\ 9 \quad 6 \quad 3 \\ 6 \quad 4 \quad 2 \\ \hline 3 \quad 2 \quad 1 \\ 9 \quad 12 \quad 10 \quad 4 \quad 1 \end{array}$$

(b)

